

Supermart Grocery Sales - Retail Analytics Dataset

What I Did in This Project

In the "Supermart Grocery Sales - Retail Analytics Dataset" project, I performed a full-cycle data analysis and basic machine learning modeling to understand retail sales patterns and derive business insights.

1. Data Loading and Preprocessing:

- Loaded the dataset using Pandas and examined data structure.
- Cleaned the data by removing null values and duplicates.
- Converted 'Order Date' to datetime format and extracted day, month, and year.

2. Data Transformation:

- Encoded categorical variables using Label Encoding.
- Created new features such as 'Order Month' and 'Order Year' for time-series analysis.

3. Exploratory Data Analysis (EDA):

- Analyzed total sales by product category and subcategory.
- Identified top-performing cities and regions using group-by operations.
- Examined monthly and yearly sales trends using line and pie charts.
- Generated a correlation heatmap to find relationships between numeric variables.

4. Machine Learning Model:

- Selected relevant features and target variable ('Sales').
- Split data into training and test sets.
- Applied feature scaling using StandardScaler.
- Built and trained a Linear Regression model to predict sales.

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- Evaluated model using Mean Squared Error and R-squared metrics.

5. Visualization:

- Created visualizations for category-wise sales, monthly trends, top cities by sales, and model predictions.

6. Conclusion and Insights:

- Identified Egg, Meat & Fish as the top sales category.
- Found that sales increased towards the end of the year.
- The regression model performed well with an R^2 score of approximately 0.82.

This project helped reinforce my skills in data cleaning, visualization, feature engineering, and applying linear regression to real-world business data.